MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSECFS201 - LINUXADMINISTRATION

(2024 ADMISSION ONWARDS)

QPCODE: 25901205

ANSWER SCHEME

Part A – Short Answer Questions

1. Associate system users with their purpose in Linux.

- **Root user:**Superuser with complete administrative privileges.
- **System users:** Used by system services and daemons.
- **Regular users:** Created for normal user activities.

2. Identify the main types of files in Linux.

- Ordinary (Regular) files
- Directory files
- Device files
- Link files (Hard and Symbolic links)
- FIFO/Pipe files
- Socket files etc

3. Indicate the role of the file system in Linux.

The file system organizes, stores, manages, and retrieves data and directories on storage devices.

4. Write a command to calculate 25 × 4 using expr.

```
expr 25 \* 4
```

5. Apply the find command to locate a file named report.txt.

```
find / -name report.txt
```

6. Define symbolic mode in chmod.

Symbolic mode uses symbols such as **u, g, o, a, +, -, =** to modify file permissions.

7. Compare the roles of cd and pwd in directory management.

- **cd:** Changes the current working directory.
- **pwd:** Displays the present working directory.

8. Define loop control structures.

Loop control structures repeatedly execute a set of commands until a specified condition is met.

9. Write a shell program to display a given string.

```
echo "Hello World"
```

10. List the keywords in shell.

Common shell keywords:

- if

- then
- else
- fi
- for
- while
- do
- done
- case
- esac

11. Describe the role of RPM in Red Hat Linux.

RPM (Red Hat Package Manager) is used to install, update, verify, and remove software packages.

12. Justify any one practical application of DNS in daily internet use.

DNS translates domain names such as **www.google.com** into IP addresses, allowing users to access websites without remembering numerical addresses.

Part B – Short Essay Questions

13. Categorize process management commands into monitoring and controlling tools.

Monitoring Commands:

- ps
- top
- pgrep

Controlling Commands:

- kill
- killall
- nice
- renice

Justification: Monitoring commands observe process status, whereas controlling commands manage or terminate processes.

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14. Explain the working of the head, tail, and wc commands.

- **head:** Displays the first 10 lines of a file.
  ```
  head file.txt
  ```
- **tail:** Displays the last 10 lines of a file.
  ```
  tail file.txt
  ```
- **wc:** Counts lines, words, and characters.
  ```
  wc file.txt
  ```

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15. Explain the purpose and usage of the ifconfig command.

The **ifconfig** command displays and configures network interfaces.
To check the IP address:

```
ifconfig
```

or

```
ifconfig eth0
```

It shows IP address, subnet mask, MAC address, and interface status.

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16. Explain mounting and unmounting file systems in Linux.

Mounting: Attaching a file system to the Linux directory tree.
Example:

```
mount /dev/sdb1 /mnt
```

Unmounting: Detaching a mounted file system.
Example:

```
umount /mnt
```

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17. Explain any four positional parameters with meanings.

- **$0** – Script name
- **$1** – First argument
- **$2** – Second argument
- **$#** – Number of arguments

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18. Describe any 5 text entry commands in vi editor.

- **i** – Insert before cursor
- **I** – Insert at beginning of line
- **a** – Append after cursor
- **A** – Append at end of line
- **o** – Open a new line below current line

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19. Justify the preference of using sudo instead of root.

- Improves security.
- Limits administrative privileges.
- Records user activities in logs.
- Reduces accidental system damage.
- Supports accountability in multi-user systems.

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20. Explain how Linux differentiates access rights between owners, groups, and others.

Linux permissions are divided into:

- **Owner (u):** Creator of the file.
- **Group (g):** Members of the assigned group.
- **Others (o):** All other users.

Permissions:

- Read (r)
- Write (w)
- Execute (x)

Example:

```
-rwxr-xr--
```

Part C – Essay Questions

21. Compare Bourne Shell, Bash Shell and C Shell.

Feature	Bourne Shell (sh)	Bash Shell	C Shell (csh)
Developer	Stephen Bourne	GNU Project	Bill Joy
Syntax	Simple	Bourne-compatible	C-like
Command History	Limited	Extensive	Available
Aliases	No	Yes	Yes
Scripting	Good	Excellent	Moderate
Popularity	Older systems	Most widely used	Academic environments

Conclusion: Bash is the most widely used shell due to its advanced features and compatibility.

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22. Explain different types of shell available in Linux with features.

Bourne Shell (sh)

- Original UNIX shell.
- Supports scripting.

Bash Shell (bash)

- Default Linux shell.
- Command history, aliases, job control.

C Shell (csh)

- C language-like syntax.
- Useful for interactive use.

Korn Shell (ksh)

- Combines features of Bourne and C Shell.
- Efficient scripting.

TC Shell (tcsh)

- Enhanced version of C Shell.
- Command completion and editing.

Z Shell (zsh)

- Powerful customization.
- Advanced completion and themes.

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23. Evaluate the effectiveness of managing users and groups separately in Linux.

Managing users and groups separately improves:

Security

- Access control based on user roles.
- Restricts unauthorized access.

Simplified Administration

- Permissions can be assigned to groups rather than individual users.
- Easier management of large organizations.

Resource Sharing

- Group members can share files and resources efficiently.

Scalability

- New users can be added to existing groups quickly.

Accountability

- User activities can be tracked individually.

Conclusion: Separate management of users and groups enhances security, flexibility, scalability, and administrative efficiency in multi-user Linux environments.